

Propellantless electron thruster for LEO station keeping

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Abstract

Small spacecraft below 600 km lose their orbit to atmospheric drag within months to a few years, and most of them carry no propulsion at all. This paper describes a thruster with no tank, no feed system and no neutralizer. A cathode held a few hundred volts below the spacecraft body emits electrons through a single aperture. The escaping beam produces thrust and leaves the body positively charged, so the body collects electrons from the ionospheric plasma until collection equals emission. The circuit closes through the ionosphere with zero net mass flow. Particle-in-cell simulations (WarpX) of a 10 mm conducting body in a dayside ionospheric plasma measure the complete operating envelope over 100–350 V: 3.4 to 43 nN of thrust from 12 to 278 mW of beam power, with 96–99 % of the beam escaping, a floating potential of 4–48 V, and an energy efficiency of 0.67–0.77. Two measured constants describe every run to within 3 %. Against a year of real 2024 drag from orbit propagation, the device cancels mean and maximum drag at 500–600 km for 7–31 mW; at 400 km it covers the mean demand with a slender body. Because both drag and electron collection scale with area, the feasibility condition is independent of vehicle size, and a 3U CubeSat needs 0.9–3.4 W for the same altitudes. A bench experiment confirms the electrode topology on hardware.

Open items are the transverse geomagnetic field, the choice of flight emitter, and the CubeSat sheath regime.

1 Introduction

Small spacecraft below 600 km lose their orbits to atmospheric drag within months to a few years. A 3U CubeSat at 400 km falls below 100 km in roughly 139 days. Most carry no propulsion at all.

The thrust to hold altitude is nanonewtons to a few micronewtons, well within the range of flight-proven micropropulsion. Colloid thrusters on LISA Pathfinder resolved 0.1 μN [1]; NanoFEEP flew on a 1U CubeSat at 0.2–0.3 W [2]. Every one of these systems spends a stored propellant through tanks, valves, feed lines or precision emitter heads. The propellant imposes a total-impulse limit, and the hardware adds mass and cost that are rarely justified on a CubeSat.

This paper describes a thruster with none of that hardware. A cathode held below the spacecraft potential emits electrons through an aperture. The escaping beam produces thrust and leaves the body positively charged, so the body collects ambient electrons from the ionospheric plasma until collection balances emission. The circuit closes through the ionosphere with no propellant and no net mass flow. An electron beam carries about 200 times less momentum per watt than an ion beam, but LEO drag on a small spacecraft is nanonewtons, and at that scale the beam costs tens of milliwatts. The whole device is a cathode, an aperture and a high-voltage supply. It carries

no propellant, no tank and no feed system, and at 400–600 km it produces enough thrust for station keeping or extending mission life.

The following sections describe the operating principle, test each claim through a ladder of particle-in-cell simulations that build from a vacuum electron gun to the full device in ionospheric plasma, and assess the result against a year of real 2024 drag at 400–600 km.

2 Concept

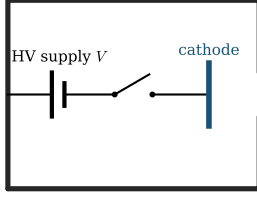
The device has three parts (Fig. 1):

1. a cathode, held at $-V$ with respect to the spacecraft body, that emits electrons and accelerates them out through an aperture;
2. the spacecraft's own conducting skin, which is the second electrode and collects electrons from the plasma;
3. a high-voltage supply between the two, which moves collected electrons back to the cathode.

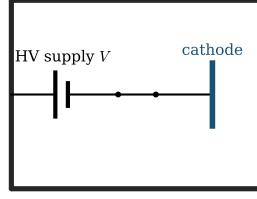
Once switched on, the device reaches a steady state:

1. the spacecraft floats in the ionospheric plasma;
2. the supply is switched on and the cathode emits;

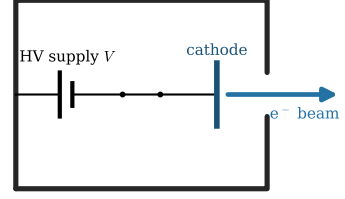
1. initial state



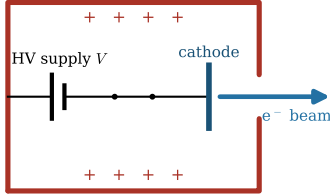
2. supply switched on



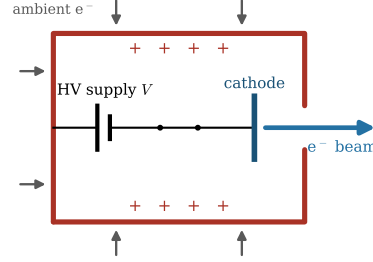
3. cathode emits, beam accelerated out



4. body charges positive



5. positive body collects ambient e-



6. steady state: collection = emission

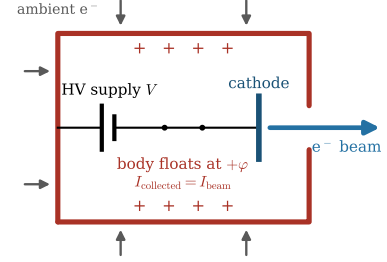


Figure 1: Operating sequence. The supply is switched on, the cathode emits, the escaping beam charges the body positive, and the positive body collects ambient electrons until collection equals emission.

3. each escaping electron leaves the body one charge more positive;
4. the positive body attracts electrons from the plasma;
5. collection grows until it equals emission. The body settles at a potential φ where the two currents balance.

Two questions decide whether this works.

Why does the positive body not recapture its own beam? Because the beam electrons carry 100–300 eV while φ is a few volts to a few tens of volts; ambient electrons at 0.1 eV are captured, beam electrons are not.

Does the returning current cancel the thrust? The collected electrons gain 4–48 eV from the floating potential, against 77–272 eV for the beam, and they arrive from all directions rather than straight astern. The simulations measure the net cancellation at 1–2 % of the beam thrust at 200–350 V, rising to 9 % at 100 V where the slower beam carries less momentum relative to the collected electrons (Sec. 4.4.1).

3 Theory

3.1 Thrust

The beam is a current I of electrons with kinetic energy KE. Its momentum flux is the thrust:

$$F = \frac{I\sqrt{2m_e \text{KE}}}{e} = 3.37 \frac{\text{nN}}{\text{mA}\sqrt{\text{eV}}} I\sqrt{\text{KE}}. \quad (1)$$

The constant contains only the electron mass and charge. The exhaust velocity is $v_e = \sqrt{2e \text{KE}/m_e}$, of order 10^7 m/s.

3.2 The float tax

The body floats at $+\varphi$ and the cathode at $\varphi - V$, so the accelerating potential is $V - \varphi$. Space charge near the cathode depresses it further. The exhaust energy is

$$\text{KE} = \kappa (V - \varphi), \quad (2)$$

where κ is the gun's energy transmission, set by how hard the gun is loaded relative to its space-charge limit. The ratio φ/V is the fraction of supply voltage spent on collecting the return current.

3.3 Power and efficiency

Beam power is $P = VI$. Jet power over supply power is

$$\eta = \kappa \frac{V - \varphi}{V} f_{\text{esc}}, \quad (3)$$

with f_{esc} the fraction of the beam that clears the body. Eliminating I between Eq. (1) and $P = VI$ at fixed thrust gives $P \propto F\sqrt{V}$: doubling the voltage costs about 40 % more power for the same thrust. Thrust per watt is $F/P \approx 3.37\sqrt{\kappa}/\sqrt{V}$ in $\mu\text{N/W}$ with V in volts, about $0.2 \mu\text{N/W}$ at 200 V.

3.4 Emission ceiling

Low voltage is cheap, but the current a gun can source is limited by space charge. The planar Child–Langmuir scale for the simulated spot ($r = 0.5$ mm, gap 4.7 mm) is $I_{\text{CL}} \propto V^{3/2}/d^2$, equal to 0.08, 0.24 and 0.43 mA at 100, 200 and 300 V; the real gun emits 1.46 times that. Maximum thrust therefore scales as $I_{\text{CL}}\sqrt{V} \propto V^2$, and at 100 V the spot sources about 0.12 mA whatever the

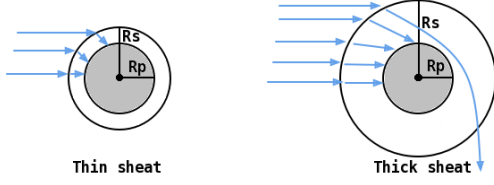


Figure 2: Thin sheath ($r_p \gg \lambda_D$): electrons reach the body on nearly straight paths and collection scales with surface area. Thick sheath ($r_p \lesssim \lambda_D$): the potential extends far beyond the body and bends electron orbits inward, giving the $(1 + \chi)^\alpha$ enhancement of Eq. (5). The 10 mm body sits in the thick-sheath regime; a CubeSat would sit in the thin-sheath regime.

plasma does. The best operating point is the lowest voltage whose emission ceiling still meets the thrust demand. A shorter gap or a wider spot moves the ceiling without changing the voltage.

3.5 Current collection

The body collects the one-sided thermal electron flux [6]

$$I_{th} = e A n_e \sqrt{\frac{kT_e}{2\pi m_e}}, \quad (4)$$

enhanced by the positive potential. With $\chi = e\phi/kT_e$,

$$I = \beta I_{th} (1 + \chi)^\alpha, \quad (5)$$

where $\alpha = 1$ for a sphere and 0.5 for a long cylinder in orbital-motion-limited theory, and β is a geometry factor. Setting I equal to the escaping current and inverting gives the float. Its stiffness goes as $1/(n_e \sqrt{T_e})$, so thin night-side plasma makes the body float higher for the same current.

3.6 Scale

Drag is proportional to the ram area; collection and body-mounted power are proportional to the skin area. Thrust demand, beam current and power all scale with ram area, so power per unit ram area depends only on altitude and voltage. Vehicle size drops out and what remains is the shape ratio A_{skin}/A_{ram} . A 10 mm can and a 3U CubeSat with the same shape ratio have the same feasibility margin.

3.7 Control

I and ϕ are plain electrical measurements, so Eqs. (1)–(2) let the spacecraft compute its own thrust in flight. Density, temperature and day/night all collapse into where the body floats, so a two-line servo on the measured float is the whole control law. A microcontroller is sufficient.

Table 1: PIC configuration of the full thruster at 200 V. Characterization runs change one entry at a time (Table 4).

	parameter	value
geometry	body outer diameter $\times$ length	10×5.5 mm
	wall thickness	0.4 mm
	exhaust aperture diameter	4 mm
	cathode disk diameter	3 mm
	emission spot diameter	1 mm
	cathode–lid gap	4.7 mm
electrical	supply voltage V	200 V
	beam current I (prescribed)	0.342 mA
	gun loading I/I_{CL}	1.46
	body self-capacitance	0.5–1 pF
plasma	n_e	$1.63 \times 10^{12} \text{ m}^{-3}$
	kT_e	114 meV
	kT_i	81 meV
	ion mass	$400 m_e$
numerics	cell size $\Delta r = \Delta z$	0.15 mm
	domain r_{max}	30 mm
	particles per cell per species	16
	run length	800 ns (29 RC times)
	supply / gun switch-on	100 / 150 ns

4 PIC simulations

4.1 Setup

All runs use WarpX [3] (built on AMReX [4]) in RZ electrostatic mode with the parameters of Table 1. The body is a closed can with a cathode disk inside and an aperture in the lid, modelled as an embedded conductor (Fig. 3). Its potential is not prescribed. Each step the code counts the charge that left through the aperture and the charge that landed on the skin, and moves the body potential by the net charge over the body’s self-capacitance, which is measured once from Gauss’s law on a vacuum Laplace solve. The float is therefore an output, and ladder stage 7 checks the method on a bare body against the theoretical floating potential. Ions are repelled by the positive body, so the balance is between the escaping beam and collected electrons only; the reduced ion mass affects how fast the sheath settles, not where it settles. The ambient plasma is a dayside 400 km row kept constant by a particle reservoir. Every run freezes its configuration and passes fixed acceptance gates on escape fraction, floating potential, current balance and momentum before it is used.

4.2 Validation ladder

A PIC simulation of the full device has too many parameters to validate at once. The ladder isolates each piece of physics and checks it against theory before combining them, so the full thruster result inherits confidence from every stage below it. Nine stages build from a vacuum electron gun to the full device (Table 2). All nine pass.

Table 2: Validation ladder.

#	stage	validates
1	negative cathode	emits through neutral aperture
2	holed anode	aperture controls transmission
3	voltage bracket	transmission is V -independent
4	thermal collection	PIC current vs Eq. (4)
5	biased, +3 V	OML collection at bias
6	biased, +10 V	larger sheath, more current
7	floating body	unbiased float vs theory
8	two-node Laplace	two potentials, one body
9	full thruster, 200 V	gun + plasma combined

Stages 1–3 test the gun alone in vacuum. Stage 1 confirms that a negative cathode emits and accelerates electrons through a near-neutral aperture, the basic electrode topology, with a $35\ \mu\text{V}$ energy error on 100 V; the same topology is confirmed on hardware in Sec. 7. Stage 2 measures aperture transmission at three hole sizes ($\kappa = 0.97, 0.90$ and 1.00 , matching the predicted values). Stage 3 confirms transmission is independent of voltage (0.006 pp spread over 200–300 V). The transmissions in stages 2–3 are for lightly loaded vacuum guns; the device gun runs at 1.46 times its space-charge scale, which is where the lower $\kappa = 0.81$ of Sec. 4.4.1 comes from.

Stages 4–7 test collection on a conducting body in the plasma. Stage 4 reproduces the thermal electron flux of Eq. (4) to within 1 %. Stages 5–6 bias the body to +3 and +10 V and measure 0.85 and 0.81 of the Laframboise OML ceiling, with the sheath growing from 4.1 to 6.9 mm as the potential rises. Stage 7 lets the body float freely and measures -0.251 V, inside the bracket between two theoretical models. Stage 8 places two prescribed potentials on one embedded conductor and recovers the exact Laplace solution with zero violation, which validates the multi-potential method used in stage 9. Stage 9 is the full thruster, described next.

4.3 Full thruster

Stage 9 puts the gun inside the floating can and switches it on in the 400 km plasma (Fig. 3). The body charges positive as beam leaves, collection rises with the float, and the two currents balance within a few RC times. Table 3 lists the steady state at 200 V. The body floats at 17 V, 98.4 % of the beam escapes, and the thrust is 13.65 nN at 68.4 mW of beam power. This run is the anchor: everything in the next section perturbs it one axis at a time.

4.4 Characterization

Eight runs vary the full thruster one axis at a time: voltage first, then shape, plasma density and magnetic field, and

Table 3: Steady-state values of the full thruster at 200 V. This run is the anchor for all characterization.

quantity	value
floating potential φ	17.0 V
beam escape f_{esc}	98.4 %
thrust F	13.65 nN
beam power P	68.4 mW
exhaust energy KE	147.5 eV
energy efficiency η	0.73

Table 4: Characterization runs. Each changes one axis of the full thruster except the last, which combines two.

axis	φ [V]	F [nN]	f_{esc} [%]	KE [eV]
full thruster (200 V)	17.0	13.65	98.4	147.5
100 V	5.4	3.42	96.1	77.2
300 V	36.3	30.13	99.0	210.1
350 V	48.3	40.48	99.1	239.0
slender body ($L/r = 6$)	4.4	14.22	98.4	159.7
density $n_e/3$	42.5	12.39	99.1	122.0
$B_z = 30\ \mu\text{T}$	17.2	13.64	98.4	147.3
$B_z = 300\ \mu\text{T}$	49.6	12.06	98.3	115.9
350 V + slender	14.0	43.33	99.1	272.7

one run that combines two axes (Table 4). All pass their gates. Predictions were written down before each run.

4.4.1 Voltage envelope

Three runs at 100, 300 and 350 V hold the same fraction of the Child–Langmuir current and the same plasma (Table 5, Fig. 4). Pre-registered thrust predictions landed within 1 %.

Two constants fit all four points:

$$F [\text{nN}] = 3.27 I [\text{mA}] \sqrt{\text{KE} [\text{eV}]}, \quad (6)$$

$$\text{KE} = 0.81 (V - \varphi). \quad (7)$$

The thrust slope is 97 % of Eq. (1); the deficit is plume divergence. $\kappa = 0.81$ is the space-charge depression of a single-aperture gun at this loading. Thrust per watt follows $1/\sqrt{V}$ within 2.5 % over 100–300 V, and the 350 V point sits 3 % below it because the float tax has grown to 14 %. The energy efficiency stays near 0.73 because almost every electron leaves at nearly full energy.

The 300 V run also selects the collection exponent. Three values of α in Eq. (5) agree at 200 V and predict 31, 36 and 90 V for the 300 V float; the measurement is 36.3 V. A fit across the envelope gives $\alpha = 0.82$ – 0.89 , between the sphere and cylinder limits. The 350 V float was predicted from that fit at 47 V and measured at 48.3 V. One caveat applies to both high-voltage points: the sheath settles on the ion timescale, and their floats were still rising slowly at run end. The quoted values are tail averages; extrapolating the decaying slope puts the settled 300 V float at 42–48 V and the settled 350 V float just past the 50 V design limit. The compact body at 350 V sits on that limit, which is why the slender body matters at 400 km.

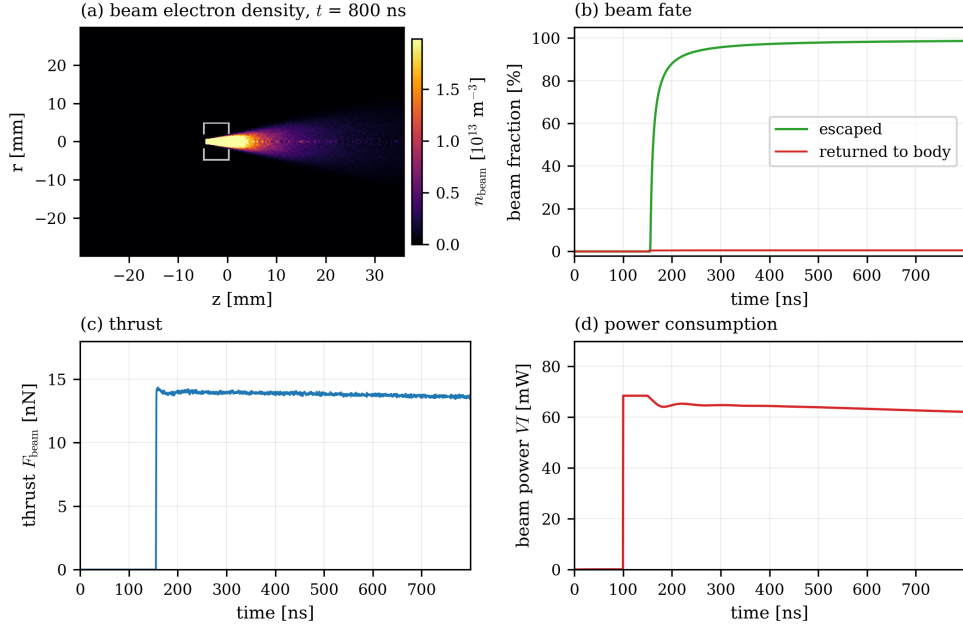


Figure 3: The full thruster at 200 V, the reference run for everything that follows. (a) Beam electron density at the end of the run; the grey outline is the can. (b) Fraction of emitted beam that has escaped the domain or returned to the body. (c) Beam thrust. (d) Beam power. The gun switches on at 150 ns; thrust, power and escape are all steady well before 800 ns.

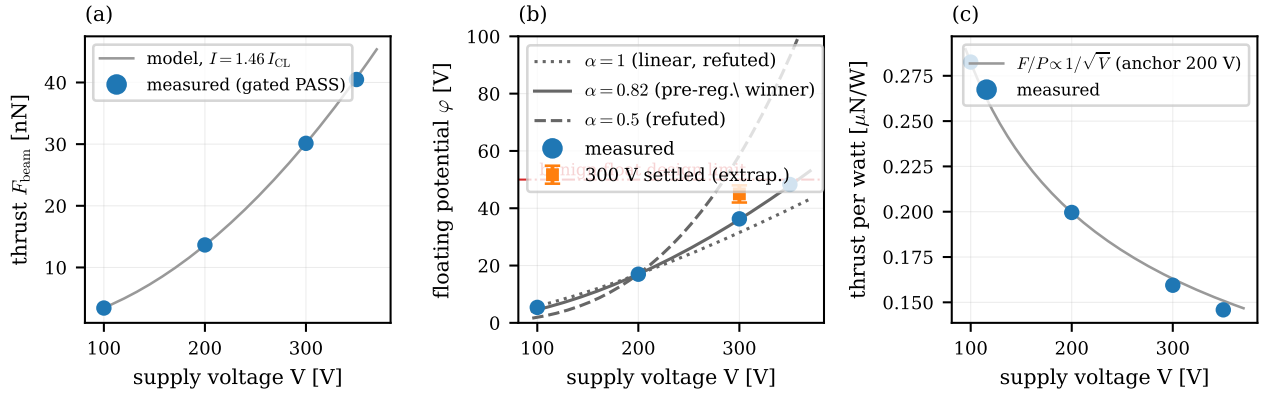


Figure 4: The four envelope runs (dots) against the model (lines). Left: measured thrust and the two-constant law $F = 3.27 I \sqrt{KE}$, $KE = 0.81(V - \phi)$. Centre: measured floating potential and the float predicted by Eq. (5) for three collection exponents, each forced through the 200 V point; they diverge above 200 V, and the 300 V measurement selects $\alpha \approx 0.82$ – 0.89 and refutes the sphere ($\alpha = 1$) and cylinder ($\alpha = 0.5$) limits. The square is the 300 V float extrapolated to sheath settlement. Right: measured thrust per watt and the $1/\sqrt{V}$ law that follows from $F \propto I \sqrt{V}$, $P = IV$, anchored at 200 V; the 350 V point falls below it because the float tax has grown.

Table 5: Measured operating envelope of the 10 mm body. Grid-resolution uncertainty: $F \pm 4\%$, $KE \pm 7\%$, $\phi \pm 2\%$.

V V	I mA	ϕ V	F nN	P mW	f_{esc} %	η
100	0.121	5.4	3.42	12.1	96.1	0.67
200	0.342	17.0	13.65	68.4	98.4	0.73
300	0.630	36.3	30.13	189	99.0	0.74
350	0.793	48.3	40.48	278	99.1	0.74

4.4.2 Shape, density and field

Shape. Does elongation hurt current collection? Stretching the can to $L/r = 6$ triples the skin at the same ram area. Two hypotheses were recorded before the run: area-only scaling predicted 4–5 V, and the long-cylinder limit predicted tens of volts. The float dropped to 4.4 V

and the thrust rose 4 %, confirming area-only scaling. The collection exponent held across a $3.2\times$ area change. A 3U CubeSat flying end-on has the same shape ratio, so the measured exponent applies directly (Sec. 5.4).

Density. Does thin night-side plasma break the device? At one third density the run was extended to $2.4\mu s$ (three times the RC time) and the float settled at 42.5 V. Every fixed-exponent prediction from Eq. (5) overshot, from 53 V at $\alpha = 1$ to 68 V at $\alpha = 0.82$. The fitted law overestimates the float in thin plasma, so night-side predictions from Sec. 5 are conservative.

Magnetic field. Does the geomagnetic field tax collection? An axial field at flight strength ($30\mu T$) changes

Table 6: Mission cases: 2024 drag on the 10 mm body and the IRI-2020 plasma along the orbit (5th–95th percentile of 5-min samples over the year).

altitude	pose	drag [nN] mean max	n_e [10^{11} m^{-3}]	kT_e [meV]
400 km	axial	32.9 92.4	2.7–21	87–193
400 km	lateral	21.6 60.7	2.7–21	87–193
500 km	axial	7.6 28.4	1.4–16	91–221
550 km	axial	3.8 16.3	1.0–11	93–233
600 km	axial	2.0 9.6	0.7–7.1	97–244

nothing: float, thrust and escape are within 0.2 V, 0.1 % and 0.06 percentage points of the reference run. At ten times flight strength, collection is taxed and the float rises 33 V; the thrust law still reproduces the result to within 1 % once the measured float is inserted. The transverse field, which is the flight geometry, remains open (Sec. 8).

400 km operating point. Does the slender body at 350 V close the 400 km mean drag (32.9 nN)? The voltage and shape laws together predicted 11–17 V and 42–43 nN. It measured 14.0 V and 43.33 nN, covering the mean demand with a float well inside the 50 V limit. The two laws compose, which turns two one-dimensional curves into an operating map over 100–350 V and $L/r = 0.6$ –6.

5 Mission assessment

5.1 Method

Drag demand comes from TudatPy [5] propagation over 2024 (solar-cycle maximum) with NRLMSISE-00 density driven by measured F10.7 and A_p , and IRI-2020 plasma along the orbit. The test body is the simulated 10 mm $\times$ 5 mm cylinder at 100 g with $C_d = 2.2$, on a circular near-equatorial orbit ($i = 0.5^\circ$) at 400 km (axial and lateral pose), 500, 550 and 600 km. Drag is cancelled exactly, so the drag at each timestep is the thrust demand. Table 6 summarizes the demand and the plasma each case sees. The PIC plasma row of Table 1 sits near the dayside top of the 400 km range; night-side densities are 5–40 times lower, which is where collection is weakest.

5.2 Coverage

At 500–600 km the thruster covers mean and maximum drag at a fixed 300 V, and in voltage-following mode it costs 7–31 mW (Fig. 5, Tables 7–8). At 400 km the picture is mixed. The compact body delivers 40.48 nN at 350 V against a 32.9 nN mean, but floats on the 50 V charging limit. The slender body delivers 43.33 nN at a 14 V float, 3.6 times below the 50 V limit, covering the mean at about 76 % duty. The drag maxima

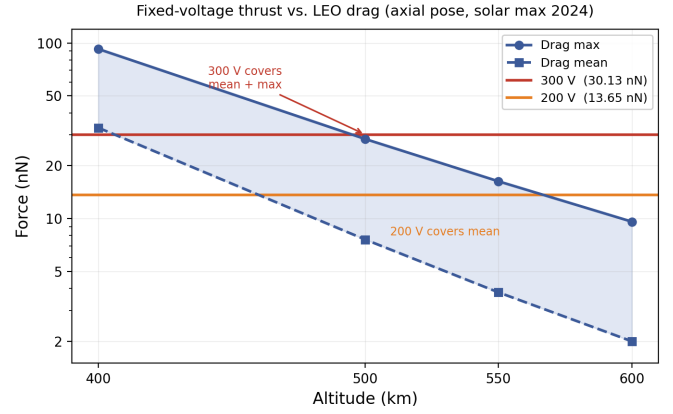


Figure 5: Measured thrust at 200 and 300 V against mean and maximum 2024 drag on the 10 mm body, by altitude.

Table 7: Fixed-voltage coverage of 2024 drag on the 10 mm body.

altitude	mean nN	max nN	200 V 13.65 nN	300 V 30.13 nN
400 km axial	32.9	92.4	–	–
400 km lateral	21.6	60.7	–	mean
500 km	7.6	28.4	mean	mean, max
550 km	3.8	16.3	mean	mean, max
600 km	2.0	9.6	mean, max	mean, max

(92 nN) exceed any single operating point, and the compact shape cannot keep up year-round because thin night-side plasma caps collection. 400 km is a research target, not a closed case. Where full station-keeping does not close, partial compensation still extends mission life.

5.3 Power

Power is the thruster’s only interface to the spacecraft: tens of milliwatts at 500–600 km, 100–200 mW at 400 km. The 10 mm body harvests roughly 10–30 mW from body-mounted cells. A 1U CubeSat has 1–2 W and a 3U 5–10 W. Supply and demand both grow with area, so the ratio holds with size.

5.4 Scaling to CubeSats

By the argument of Sec. 3, a 3U CubeSat flying end-on has the same shape ratio as the slender can and the same feasibility margin. In absolute terms it needs 8.8 mA and 0.9 W at 600 km, 1.7 W at 550 km and 3.4 W at 500 km. Larger bodies also need less collection enhancement over the thermal flux (4 times for a 3U at 600 km against 15 times for the 10 mm body), so they rest on less extrapolation, not more. These numbers are estimates from the fitted collection law, and CubeSat radii sit at 25–60 Debye lengths where that law has not been measured (Sec. 8).

Table 8: Voltage-following mode: the lowest voltage that cancels each timestep’s drag, and the resulting mean power.

altitude	$V_{\min}$	mean P	in envelope
600 km	100 V	6.8 mW	yes
550 km	104 V	13.4 mW	yes
500 km	146 V	31.3 mW	yes
400 km lateral	247 V	116 mW	yes
400 km axial	304 V	196 mW	yes (350 V)

6 Comparison with micropropulsion

Table 9 compares the device with flown systems in the nanonewton-to-micronewton range. The thrust per watt is about 200 times below an ion thruster, which is Eq. (1) and cannot be engineered away. At micronewtons the power penalty reaches watts and electrospray or FEEP wins. The difference from every flown system is the hardware column: no tank, no feed system, no neutralizer, no total-impulse limit. Nothing in the table produces thrust below 100 nN.

Related concepts share the return-circuit idea. Bare electrodynamic tethers [6, 7] and Janhunen’s electric sail and plasma brake [8, 9] all use continuous electron emission with the ambient plasma closing the circuit, but produce thrust by Lorentz force or ion deflection on kilometre-scale tethers. Atmosphere-breathing electric propulsion [10] collects ambient gas but keeps an ionization stage, an acceleration stage and a neutralizer. A recent review of propellantless propulsion [11] does not list a standalone ionospheric electron-exchange thruster.

7 Bench experiment

The electrode topology was tested in a laboratory vacuum chamber. A W5W automotive bulb filament with the glass removed, held at -50 and -62 V on its two terminals, faces a grounded 4×4 cm copper plate in a glass chamber at $4\text{--}5$ Pa. The ammeter between plate and earth reads a steady 87 mA once the filament is hot: a single net-negative electrode emits electrons and accelerates them across a gap onto a grounded collector. At this pressure the gap is collisional, so the current includes gas amplification and the result is qualitative. The experiment confirms the principle tested numerically in ladder stage 1: a negative cathode emits and accelerates electrons toward a grounded plate. Controls (bias off, filament cold) and an I–V sweep are the next step.

8 Limitations

Ordered by how much each could change the answer.

1. **Transverse geomagnetic field.** In LEO the field is roughly perpendicular to the thrust axis. The beam

gyroradius (1.4 m) is 50 times the simulation domain, so the fate of the exhaust momentum once the beam gyrates is unmeasured and could move the thrust either way.

2. **Prescribed beam.** Beam current is a fixed input, not self-consistently emitted from a modelled cathode. The choice of flight emitter is outside the scope of this paper.
3. **CubeSat sheath regime.** Runs sit at $r/\lambda_D \approx 2.5$; CubeSats are $25\text{--}60 \lambda_D$ across, where orbital-motion-limited collection no longer applies. A single large-body run would settle this.
4. **Reduced ion mass, stationary plasma, held attitude.** Ions are $400 m_e$, no ram flow is modelled, and the mission cases assume a fixed pose without an attitude-control model.

All powers are beam power VI ; converter and cathode overheads belong to the flight design.

9 Conclusion

A cathode, a high-voltage supply and the spacecraft’s own skin make a thruster that needs no propellant. PIC simulations measure its operating envelope at $3\text{--}43$ nN from $12\text{--}278$ mW, with two constants describing every run to within 3 %, a floating potential fixed by a measured collection law, and a return current that costs 1–9 % of the beam thrust. Against a year of real drag the device holds $500\text{--}600$ km for tens of milliwatts and covers the 400 km mean demand with a slender body. The feasibility condition depends on shape and altitude but not on size, so a 3U CubeSat inherits it at $0.9\text{--}3.4$ W. The device is cheap to test: it needs no hardware beyond a supply and an electron gun of the kind that already flies as a neutralizer, and it can be retrofitted to an existing ion or Hall thruster after its propellant is spent. The open questions, the transverse field and the flight emitter, are the next work.

Data. All simulation inputs, run manifests, gate policies, the orbit propagations and the model are committed in the project repository.

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Table 9: Position against flight micropropulsion in the nanonewton-to-micronewton range.

system	example	thrust	power	I_{sp} [s]	hardware
this work	electron exchange	3–43 nN	12–278 mW	10^6	HV supply, cathode, aperture
cold gas	GomSpace	0.1–10 mN	<2 W	60–110	tanks, valves, nozzles
PPT	NanoProp [12] PPTCUP [13]	40 μ N	2 W	655	capacitor bank, electrodes
electrospray	Busek CMNT [12]	5–30 μ N	1–16 W	200–300	tanks, PPU, neutralizer
FEEP (1U)	NanoFEEP [2]	0–8 μ N	0.2–0.3 W	6000	liquid-metal needles, heater
FEEP	Enpulsion IFM Nano [14]	10–500 μ N	8–40 W	2000–5000	emitter, 2 neutralizers, PPU

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